

E-Commerce Portal PRD

Product Overview

Build a modern e-commerce platform consisting of:

1. Customer-facing storefront
2. Administrative dashboard
3. Product catalog management
4. Inventory management
5. Order management

The platform should support complete product discovery, purchasing, fulfillment, and operational workflows.

Objectives

Business Goals

- Enable online product sales
- Simplify inventory operations
- Centralize order management
- Provide operational visibility
- Support future scalability

User Goals

Customers

- Discover products easily
- Purchase products efficiently
- Track order progress
- Manage personal information

Administrators

- Manage products and categories
 - Monitor inventory levels
 - Process customer orders
 - Access business insights
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User Roles

Customer

Can:

- Register account
- Login/logout
- Browse products
- Search products
- Manage cart
- Place orders
- Track orders
- Manage profile

Admin

Can:

- Manage products
- Manage categories
- Manage inventory
- Manage orders
- Manage customers
- View analytics

Inventory Manager

Can:

- Manage stock
- View inventory reports
- Process stock adjustments

Functional Requirements

Authentication

Customer Authentication

- Registration
- Login
- Logout
- Password reset
- Profile management

Admin Authentication

- Secure login
 - Role-based permissions
 - Session management
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Product Catalog

Product Management

Admin must be able to:

- Create products
- Update products
- Archive products
- Activate/deactivate products

Product Information

Each product should support:

- Name
- SKU
- Description
- Price
- Sale price
- Images
- Category
- Brand
- Stock quantity
- Status

Product Discovery

Customers must be able to:

- Search products
- Filter products
- Sort products
- Browse categories

Product Details

Display:

- Images
- Description
- Pricing
- Availability
- Related products

Category Management

Admin can:

- Create categories
- Update categories
- Delete categories
- Organize hierarchical categories

Shopping Cart

Customers can:

- Add products
- Remove products
- Update quantities
- View totals
- Save cart state

Cart calculations must include:

- Subtotal
- Discounts
- Taxes
- Shipping
- Grand total

Checkout

Shipping Information

Customer provides:

- Full name
- Address
- City
- State
- Country
- Postal code

Order Review

Display:

- Products
- Quantities
- Pricing

- Taxes
- Shipping charges
- Final amount

Order Placement

Customer confirms purchase and creates order.

Order Management

Customer Features

View:

- Order history
- Order details
- Order status

Order Status Flow

- Pending
- Confirmed
- Processing
- Shipped
- Delivered
- Cancelled
- Refunded

Admin Features

- View orders
 - Update status
 - Process refunds
 - View order history
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Inventory Management

Inventory Tracking

Track:

- Available stock
- Reserved stock
- Inventory movements

Inventory Operations

- Stock addition
- Stock deduction

- Inventory adjustment

Inventory Alerts

System should notify administrators when inventory falls below configured thresholds.

Customer Management

Admin should be able to:

- View customers
 - View order history
 - View customer activity
 - View customer spending
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Dashboard & Analytics

Provide visibility into:

Sales

- Revenue
- Orders
- Average order value
- Conversion metrics

Inventory

- Low stock products
- Out of stock products
- Inventory valuation

Products

- Best sellers
- Category performance

Customers

- New customers
 - Returning customers
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Notifications

Customer Notifications

- Registration confirmation

- Order confirmation
- Shipping updates
- Delivery updates

Admin Notifications

- New orders
 - Low stock alerts
 - Refund requests
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Non-Functional Requirements

Performance

- Fast page loads
- Responsive UI
- Optimized API responses

Security

- Secure authentication
- Authorization controls
- Protected admin access
- Input validation
- Audit logging

Scalability

System should support:

- Large product catalogs
- Growing customer base
- Increasing order volume

Reliability

- Error handling
- Monitoring
- Logging
- Recovery mechanisms

Accessibility

- Keyboard navigation
 - Screen reader support
 - WCAG compliance
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Future Enhancements

- Payment gateways
- Product reviews
- Wishlist
- Coupons
- Promotions
- Loyalty program
- Recommendations
- Multi-warehouse inventory
- Multi-vendor marketplace
- AI-powered search